

# VERTEX INSTITUTE OF TECHNOLOGY | Department of Applied Physics Laser Diode Dynamics, Interferometry & TIA Circuit Record

Course Code: **PHYS-402** | Module: Electro-Optic Measurement Systems

## EXPERIMENT RECORD

# **EXP-2026-04**

Status: **VERIFIED & PASSED**

**Lead Exp.:** Elena Rostova  
**Lab Partner:** Julian Thorne

**Date:** 2026-07-24  
**Supervisor:** Dr. Vance

**Station:** Bench B-07  
**Laser Class:** Class 3B  
 (650nm)

**Ambient Temp:** 21.4 °C  
**Rel. Humidity:** 44 %

## I. Executive Summary & Experimental Objectives

This laboratory investigation presents a complete electro-optical characterization of a Distributed Feedback (DFB) Semiconductor Laser Diode ( $\lambda_0 = 650$  nm), an ultra-sensitive Michelson Interferometric Displacement System, and a wideband Photodiode Transimpedance Amplifier (TIA) signal conditioning stage.

### Key Experimental Objectives

1. Measure  $I - I - V$  characteristic curves to determine threshold current  $I_{th}$  and slope efficiency  $\eta_d$ .
2. Quantify fringe visibility  $V$  as a function of optical path difference  $\Delta L$  in a Michelson setup.
3. Optimize a high-speed TIA circuit for gain-bandwidth ( $f_{-3dB} \geq 12$  MHz) with phase margin  $\phi_m \geq 60^\circ$ .

### Calibrated Test Instruments

| Device       | Model    | S/N     | Uncert.      |
|--------------|----------|---------|--------------|
| Power Meter  | ThorLabs | OP-9842 | $\pm 0.8\%$  |
| Oscilloscope | Keysight | MY5412  | $\pm 0.5\%$  |
| Piezo Ctrl   | ThorPZT  | PZ-4410 | $\pm 1.2$ nm |
| SourceMeter  | Keithley | SM-8821 | $\pm 0.05\%$ |

### Safety Precautions & ESD Operating Rules:

1. Class 3B laser radiation hazard ( $P_{max} = 15$  mW at 650 nm). Alignment must be performed strictly using certified OD4+ laser safety eyewear.
2. PIN photodiode and OPA847 op-amp inputs are extremely sensitive to electrostatic discharge ( $V_{ESD,max} = 250$  V). Wrist grounding straps were worn throughout all circuit assemblies.

## II. Theoretical Foundations & Governing Equations

The optical output power  $P_{opt}$  of the semiconductor laser diode above threshold current  $I_{th}$  is governed by the differential quantum efficiency  $\eta_d$ :

$$P_{opt}(I) = \eta_d \frac{\hbar\omega}{q} (I - I_{th}) \quad \text{for } I > I_{th} \quad (1)$$

where  $\hbar\omega = \frac{hc}{\lambda_0}$  represents the photon energy and  $q = 1.602 \times 10^{-19}$  C is the elementary electronic charge.

In the Michelson interferometer configuration, the superposition of the reference beam  $E_1$  and signal beam  $E_2$  yields a total detected optical intensity  $I_{det}(\delta)$ :

$$I_{det}(\delta) = I_1 + I_2 + 2\sqrt{I_1 I_2} \gamma(\tau) \cos\left(\frac{4\pi\Delta d}{\lambda_0} + \phi_0\right) \quad (2)$$

where  $\Delta d$  is the mirror displacement,  $\gamma(\tau)$  is the complex degree of temporal coherence, and the interference fringe visibility  $V$  is defined as:

$$V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} |\gamma(\tau)| \quad (3)$$

For the front-end optical detection circuit, the photocurrent  $I_{pd} = \mathcal{R} \cdot P_{opt}$  (where  $\mathcal{R} = 0.45$  A/W is photodiode responsivity) is converted to output voltage  $V_{out}$  via a transimpedance amplifier. The  $-3$  dB bandwidth  $f_{-3dB}$  and optimal feedback capacitance  $C_f$  to prevent peaking are given by Butterworth filter criteria:

$$C_f = \sqrt{\frac{C_{in}}{2\pi R_f f_T}}, \quad f_{-3dB} = \sqrt{\frac{f_T}{2\pi R_f C_{in}}} \quad (4)$$

where  $C_{in} = C_{pd} + C_{diff} + C_{cm}$  is the total input capacitance and  $f_T$  is the Gain-Bandwidth Product (GBWP) of the operational amplifier.